# MUAN MEURER

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## EDUCATION

**Carnegie Mellon University -** Pittsburgh, PA **August 2025 - December 2026 (expected)**
Master of Science in Mechanical Engineering | GPA 3.89/4.00
Relevant Coursework: *Robot Dynamics, Applied FEA, Space Robotics, Mobile Robots*, *Electromechanical Systems Design*

**California State University, Los Angeles** – Los Angeles, CA **August 2017 - May 2022**
Bachelor of Science in Physics | The Early Entrance Program | GPA 3.86/4.00

## PROJECTS

**NASA Lunabotics - CMU |** *Project Lead & Mechanical Engineer* **September 2025 - Present**

- Led a 73-person multidisciplinary team through the end-to-end development of an autonomous lunar excavation rover, spanning chassis, drivetrain, excavation, sensing, and avionics integration.
- Authored system-level requirements, ICDs, and risk/verification plans following SRR/PDR/CDR review structure.
- Designed and analyzed load-bearing structures for excavation and mobility in abrasive lunar regolith, accounting for traction limits and duty-cycle loading.
- Performed FEA-driven design iteration (ANSYS) to test stress, deflection, and safety margins before fabrication.
- Drove integration with electrical and software teams, resolving mechanical-avionics constraints around sensors, harnessing, and serviceability.

**Automated Coil Winder - Westinghouse & CMU |** Mechatronics Engineer **January 2026 - Present**

- Designed and built an automated variable-diameter helical coil winder for Westinghouse's eVinci™ microreactor team to prototype high-fidelity LVDT sensor coils.
- Developed a lathe-style mechatronic winding system with spindle rotation, synchronized linear wire traversal, passive tensioning, and interchangeable mandrels for repeatable coil fabrication.
- Collaborated weekly with Westinghouse engineers to document design decisions, validate test results, and deliver a functional prototype capable of producing inductance coils with less than 3% inductance variability.

**Moonranger - CMU |** *Lead Mechanical Engineer, Thermal Engineer* **August 2025 - Present**

- Led design and verification of the rover wheel, camera shroud, and deployment faux-rover using modal and structural FEA to evaluate launch-load response, joint stresses, and mounting requirements.
- Developed fabrication-ready CAD and shop drawings, coordinated vendor quotes and procurement, and managed hardware delivery schedules for wheel, shroud, and faux-rover assemblies.
- Built 1500+ node Thermal Desktop models to evaluate lunar rover thermal survivability across transit heat cycles.
- Modeled conduction, radiation, and internal dissipation to assess component stress under worst-case hot/cold cases.
- Coupled thermal, mechanical, and battery models to ensure system viability over a 14-day lunar operation window.

**The Nerenberg Lab – CSULA |** *Biophysics Research Student* **November 2021 - May 2022**

- Ran large HPC simulations (10M+ timesteps) to simulate numerically stable/convergent protein folding.
- Designed and validated simulation pipelines, stress-testing assumptions, and documented failure modes.
- Authored a 60-page technical thesis detailing methodology robustness, uncertainty sources, and reproducibility.

## PROFESSIONAL EXPERIENCE

**V3 Electric |** *Solar Salesman* **July 2024–July 2025**

- Developed working knowledge of solar energy systems, translating technical specs into customer-facing designs.
- Closed $300K/month in revenue, demonstrating disciplined execution and outstanding stakeholder communication.

## LEADERSHIP

**The Early Entrance Program Club |** *President* **May 2021 - May 2022**

- Led a 20-member executive council, managed a $10K+ budget, and coordinated programming for 100 participants.

## SKILLS

**Mechanical & Hardware:** SolidWorks, Fusion 360, GD&T, DFM, structural design, drivetrain systems, rapid prototyping
**Analysis & Simulation:** ANSYS Mechanical/Workbench, FEA, Thermal Desktop, safety margins, trade studies
**Programming & Controls:** Python, Path Planning Algorithms, MATLAB, C++, Arduino, PID motor control